Aisha Surve

AI Engineer (Intern) — LLM Applications: RAG · Agents · Evaluation

Dubai, UAE · aishasurve2007@gmail.com · +971 521739397 · github.com/aishasurve2007 · https://huggingface.co/spaces/AishaSurve

SUMMARY

Computer Science (AI) undergraduate who builds LLM-powered applications — RAG systems, tool-calling agents, and AI features served through production backends. Ships complete, tested systems and measures them honestly: benchmarks against baselines, detects data leakage, and reports real failure rates rather than best-case accuracy.

SKILLS

- **AI / ML:** RAG pipelines, LLM tool-calling agents, FAISS, BM25, rerankers, NLI hallucination detection, evaluation harnesses, faster-whisper, tree-sitter, scikit-learn, pandas, NumPy
- **Backend / Web:** FastAPI, SQLAlchemy 2.0, async jobs, REST APIs, Django, Streamlit, HTML/CSS, JavaScript
- **Languages & Tools:** Python, Java, SQL · Git, GitHub Actions (CI), pytest, Docker, MCP, FFmpeg, MySQL, VS Code

PROJECTS

VisionCut AI — Agentic video-highlight pipeline · code · live demo

- Built an end-to-end system: FastAPI async job backend → faster-whisper transcription → a GPT tool-calling agent that ranks highlights → FFmpeg cutting → downloadable clips; also exposed over an MCP server so an external client (e.g. Claude) can orchestrate the pipeline.
- Grounded the agent's picks to real transcript segments (dropping any hallucinated index) so a highlight can never point at an invented timestamp; separated LLM selection from deterministic cutting. 18 tests + CI. *Python · FastAPI · SQLAlchemy · OpenAI · MCP · FFmpeg*

StudyMate AI — RAG document Q&A with evaluation harness · code · live demo

- Built a retrieval-augmented Q&A app with hybrid retrieval and NLI-based hallucination detection; tuned with a 30-question evaluation set (locked a retrieve-20 / top-5 configuration) and deployed to Hugging Face Spaces. *Python · FAISS · BM25 · rerankers · Streamlit*

UAC Care Capacity — Honest demand nowcaster · code · live demo

- Detected target leakage that had inflated an initial model to R^2=0.996; rebuilt it as an honest nowcaster with walk-forward validation and a documented post-mortem, benchmarking against a persistence baseline (MAE 10.27 vs. linear-regression MAE 9.34). *Python · scikit-learn · Streamlit*

Codebase Intelligence Agent — Code-aware retrieval + test generation · code · live demo

- Built from scratch: tree-sitter AST chunking, hybrid BM25 + vector retrieval, and a tool-calling agent that generates pytest tests; evaluated with a 20-question harness (90% file accuracy, 75% citation accuracy). *Python*

EXPERIENCE

AI & Web Development Intern — Plugin Live · code · deployed demo *[Nov 2025 – Apr 2026]*

- Developed an AI-powered interview platform with NLP resume analysis and automated question generation (chatbot + structured modes).
- Built candidate evaluation using speech / answer-quality analysis and computer-vision behavioral insights, plus automated reporting and a secure QR-based attendance system. *Python · Node.js · Angular*

Data Science Intern — Unified Mentor · code · live demo *[May 2026 – Aug 2026]*

- Smart-factory efficiency analysis on Industrial IoT sensor data (~100k records): EDA, machine-health diagnostics, and a Random Forest classifier, with a multi-tab Streamlit monitoring dashboard (KPI tracking, alerting).
- Built course-demand and revenue forecasting models (EduPro) with feature engineering and interactive dashboards to support planning decisions. *Python · scikit-learn · Streamlit*

EDUCATION

BSc Computer Science with Artificial Intelligence — Heriot-Watt University *2025 – Present*

Higher Secondary (PCM + Computer Science) — RN Podar Senior Secondary *2023 – 2025*

CERTIFICATIONS

- Complete A.I., Machine Learning & Data Science Bootcamp (Udemy) · 100 Days of Code: Python Pro Bootcamp (Udemy)